

Exposé XII

NIELSEN–WECKEN AND LEFSCHETZ FORMULAS IN ALGEBRAIC GEOMETRY

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1.

Let X' be a scheme proper over the algebraically closed field k , and let G be a finite group acting on X' .

For each element $g \in G$, denote by

$$g_{X'} : X' \rightarrow X'$$

the automorphism of X' induced by g .

Suppose that there exists an open subset U' of X' , stable under G , such that G acts freely on U' , and let

$$\Lambda_n = \mathbb{Z}/\ell^n \mathbb{Z},$$

where ℓ is a prime number different from the characteristic of k .

Let $\Lambda_{n,U',X'}$ denote the sheaf $i_!(\Lambda_{n,U'})$ on X' , where $\Lambda_{n,U'}$ is the constant sheaf on U' associated with the ring Λ_n and where $i : U' \hookrightarrow X'$ is the canonical immersion of U' into X' . The sheaf $\Lambda_{n,U',X'}$ is endowed in a natural way with the structure of a sheaf with group G of operators. Consequently the complex

$$H_{n,U',X'} = R\Gamma_{X'}(\Lambda_{n,U',X'})$$

is a complex of $\Lambda_n[G]$ -modules, and more precisely a perfect complex (VIII, 7) of $\Lambda_n[G]$ -modules (IX, 2).

We shall see below that it is more convenient for our purpose to consider, instead of the complex $R\Gamma_{X'}(\Lambda_{n,U',X'})$, its dual

$$H_{n,U',X'}^\vee = R\Gamma_{X'}(\Lambda_{n,U',X'})^\vee$$

with respect to the ring Λ_n , in the sense of derived categories:

$$H_{n,U',X'}^\vee = R\Gamma_{X'}(\Lambda_{n,U',X'})^\vee = R\mathrm{Hom}_{\Lambda_n}(R\Gamma_{X'}(\Lambda_{n,U',X'}), \Lambda_n).$$

The complex $R\Gamma_{X'}(\Lambda_{n,U',X'})^\vee$ is a perfect complex of $\Lambda_n[G]$ -modules.

Suppose now that an endomorphism $f' : X' \rightarrow X'$ of the scheme X' and an endomorphism $\varphi : G \rightarrow G$ of the group G are given, satisfying the following two conditions:

$$f'g_{X'} = \varphi(g)_{X'}f' \quad \forall g \in G. \tag{1.1}$$

$$\text{The complement } Y' = X' - U' \text{ is stable under } f'. \tag{1.2}$$

Under the hypotheses already made, Y' is stable under G . In general one does not require the open subset U' to be stable under f' .

We therefore obtain homomorphisms of sheaves

$$\begin{aligned} f'_{\Lambda_{n,X'}} : \Lambda_{n,X'} &\longrightarrow \Lambda_{n,X'}, & f'_{\Lambda_{n,Y'}} : \Lambda_{n,Y'} &\longrightarrow \Lambda_{n,Y'}, \\ f'_{\Lambda_{n,U',X'}} : \Lambda_{n,U',X'} &\longrightarrow \Lambda_{n,U',X'}, & g_{\Lambda_{n,U',X'}} : \Lambda_{n,U',X'} &\longrightarrow \Lambda_{n,U',X'}, \end{aligned}$$

such that the following diagram, whose rows are exact, is commutative:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \Lambda_{n,U',X'} & \longrightarrow & \Lambda_{n,X'} & \longrightarrow & \Lambda_{n,Y'} & \longrightarrow & 0 \\ & & \downarrow f'_{\Lambda_{n,U',X'}} & & \downarrow f'_{\Lambda_{n,X'}} & & \downarrow f'_{\Lambda_{n,Y'}} & & \\ 0 & \longrightarrow & \Lambda_{n,U',X'} & \longrightarrow & \Lambda_{n,X'} & \longrightarrow & \Lambda_{n,Y'} & \longrightarrow & 0. \end{array} \quad (1.3)$$

We deduce in particular homomorphisms of complexes of Λ_n -modules

$$\begin{aligned} f'_{H_{n,U',X'}} &= R\Gamma_{X'}(f'_{\Lambda_{n,U',X'}}) : H_{n,U',X'} \rightarrow H_{n,U',X'}, \\ f'^{\vee}_{H_{n,U',X'}} &= (f'_{H_{n,U'}})^{\vee} : H_{n,U',X'}^{\vee} \rightarrow H_{n,U',X'}^{\vee}, \\ g_{H_{n,U',X'}} &= R\Gamma_{X'}(g_{\Lambda_{n,U',X'}}) : H_{n,U',X'} \rightarrow H_{n,U',X'}, \\ g^{\vee}_{H_{n,U',X'}} &= (g_{H_{n,U'}})^{\vee} : H_{n,U',X'}^{\vee} \rightarrow H_{n,U',X'}^{\vee}, \end{aligned}$$

such that the following relations hold:

$$g_{H_{n,U',X'}} \circ f'_{H_{n,U',X'}} = f'_{H_{n,U',X'}} \circ \varphi(g)_{H_{n,U',X'}}, \quad (1.4)$$

$$f'^{\vee}_{H_{n,U',X'}} \circ g^{\vee}_{H_{n,U',X'}} = \varphi(g)^{\vee}_{H_{n,U',X'}} \circ f'^{\vee}_{H_{n,U',X'}}. \quad (1.5)$$

Relation (1.5) shows (cf. Exposé XI on the general theory of traces, 5.2.1, 7.1) that the trace

$$\mathrm{Tr}_{\Lambda_n[G],\varphi}^*(f'^{\vee}_{H_{n,U',X'}}) \in \Lambda_n[G]_{\natural,\varphi} \simeq \mathcal{F}_!(G_{\natural,\varphi}, \Lambda_n)$$

is defined.

Our first aim will be to give, under some additional hypotheses, an explicit formula for $\mathrm{Tr}_{\Lambda_n[G],\varphi}^*(f'^{\vee}_{H_{n,U',X'}})$ in terms of local invariants associated with f' , terms which we shall define below.

2.

In fact, one can do better. For this, let n, m be two positive integers such that $m \geq n$. The ring homomorphisms

$$\alpha_{m,n} : \Lambda_m \longrightarrow \Lambda_n$$

give rise to homomorphisms of sheaves of rings on X' :

$$\begin{aligned} \alpha_{m,n,X'} : \Lambda_{m,X'} &\longrightarrow \Lambda_{n,X'}, \\ \alpha_{m,n,U',X'} : \Lambda_{m,U',X'} &\longrightarrow \Lambda_{n,U',X'}. \end{aligned}$$

Application of the Künneth formula gives

$$R\Gamma_{X'}(\Lambda_{m,X'}) \otimes_{\Lambda_m}^{\mathbb{L}} \Lambda_n \simeq R\Gamma_{X'}(\Lambda_{n,X'}), \quad (2.1)$$

$$R\Gamma_{X'}(\Lambda_{m,U',X'}) \otimes_{\Lambda_m}^{\mathbb{L}} \Lambda_n \simeq R\Gamma_{X'}(\Lambda_{n,U',X'}), \quad (2.2)$$

$$R\Gamma_{X'}(f'_{\Lambda_{m,X'}}) \otimes_{\Lambda_m}^{\mathbb{L}} \Lambda_n \simeq R\Gamma_{X'}(f'_{\Lambda_{n,X'}}), \quad (2.3)$$

$$R\Gamma_{X'}(f'_{\Lambda_{m,U',X'}}) \otimes_{\Lambda_m}^{\mathbb{L}} \Lambda_n \simeq R\Gamma_{X'}(f'_{\Lambda_{n,U',X'}}). \quad (2.4)$$

One deduces easily that

$$R\mathrm{Hom}_{\Lambda_m}(R\Gamma_{X'}(\Lambda_{m,U',X'}), \Lambda_m) \otimes_{\Lambda_m}^{\mathbb{L}} \Lambda_n \simeq R\mathrm{Hom}_{\Lambda_n}(R\Gamma_{X'}(\Lambda_{n,U',X'}), \Lambda_n), \quad (2.5)$$

$$f'_{H_{m,U',X'}} \otimes_{\Lambda_m}^{\mathbb{L}} \Lambda_n \simeq f'_{H_{n,U',X'}}. \quad (2.6)$$

The morphism of perfect complexes $R\Gamma_{X'}(f'_{\Lambda_{n,X'}})$ (resp. $R\Gamma_{X'}(f'_{\Lambda_{n,U',X'}})$, $f'_{H_{n,U',X'}}^{\vee}$) is therefore obtained by extension of scalars (XI, 6.3) from the morphisms $R\Gamma_{X'}(f'_{\Lambda_{m,X'}})$ (resp. $R\Gamma_{X'}(f'_{\Lambda_{m,U',X'}})$, $f'_{H_{m,U',X'}}^{\vee}$) by means of the morphisms $\alpha_{m,n}$ (resp. $\alpha_{m,n}$, $\alpha_{m,n,G} : \Lambda_m[G] \rightarrow \Lambda_n[G]$).

Using (XI, 6.3, 6.3.2, 6.3.3), 2.3, and 2.6, one obtains the relations

$$\mathrm{Tr}_{\Lambda_n}^*(R\Gamma_{X'}(f'_{\Lambda_{n,X'}})) = \alpha_{m,n}(\mathrm{Tr}_{\Lambda_m}^*(R\Gamma_{X'}(f'_{\Lambda_{m,X'}}))), \quad (2.7)$$

$$\mathrm{Tr}_{\Lambda_n[G],\varphi}^*(f'_{H_{n,U',X'}}^{\vee}) = \alpha_{m,n,G}(\mathrm{Tr}_{\Lambda_m[G],\varphi}^*(f'_{H_{m,U',X'}}^{\vee})). \quad (2.8)$$

The system of functions

$$(\mathrm{Tr}_{\Lambda_n[G],\varphi}(f'_{H_{n,U',X'}}^{\vee}))_{n>0}$$

defines a function

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G],\varphi}(f'_{H_{U',X'}}^{\vee}(\ell)) \in \mathcal{F}_!(G_{\mathfrak{q}\varphi}, \mathbb{Z}_\ell),$$

which is φ -central on G and has values in the ring $\mathbb{Z}_\ell$ of ℓ -adic integers.

Similarly, the elements $\mathrm{Tr}_{\Lambda_n}^*(R\Gamma_{X'}(f'_{\Lambda_{n,X'}}))$, $n \geq 1$, define an ℓ -adic integer

$$\mathrm{Tr}_{\mathbb{Z}_\ell}^*(f'_{R\Gamma_{X'}(\mathbb{Z}_\ell,X')}).$$

Our aim will be to find a formula for

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G],\varphi}^*(f'_{H_{U',X'}}^{\vee}(\ell)).$$

3. The local Nielsen–Wecken invariant (cf. [1])

Suppose, in addition to the hypotheses described in number 1, that every orbit of G is contained in an affine open subset. Then a quotient $X = X'/G$ exists and, by formula 1.1, the endomorphism f' of X' descends to the quotient:

$$f : X \longrightarrow X.$$

We shall use the following notation:

$$\begin{aligned} p : X' &\longrightarrow X = X'/G \quad \text{for the canonical projection,} \\ Y &= p(Y') = Y'/G, \\ U &= X - Y. \end{aligned}$$

The open U' is a principal covering with structural group G . Let $x \in U$ be a fixed point of f . We shall associate with it a φ -conjugacy class (XI, 7.1), denoted $\text{NW}_f^{G,\varphi}(x)$, as follows.

Let $x' \in U'$ be such that $p(x') = x$. Since U' is a principal covering with structural group G and since x is fixed by f , there exists a unique element $g \in G$ such that $f'(x') = gx'$. The φ -conjugacy class of g depends only on x . Indeed, if $p(x'') = x$, there exists $h \in G$ such that $hx'' = x'$. Consequently, if $f'(x'') = g'x''$, then

$$\begin{aligned} gx' &= f'(x') = f'(hx'') = \varphi(h)f'(x'') = (\varphi(h)g'h^{-1})x', \\ g &= \varphi(h)g'h^{-1}, \end{aligned}$$

that is, g and g' are φ -conjugate. Thus one obtains a well-defined φ -conjugacy class, the local Nielsen–Wecken invariant associated with the fixed point $x \in U$ of f , which will be denoted by $\text{NW}_f^{G,\varphi}(x)$.

The φ -conjugacy class $\text{NW}_f^{G,\varphi}(x)$ defines a φ -central function (cf. XI, 7.1)

$$\text{NW}_f^{G,\varphi}(x) : G \longrightarrow \mathbb{Z}$$

with integer values, characterized by the following relations:

$$\text{NW}_f^{G,\varphi}(x)(g) = \begin{cases} 1 & \text{if there exists } x' \in X' \text{ such that } (g^{-1}f')(x') = x' \\ & \text{and } p(x') = x, \\ 0 & \text{otherwise.} \end{cases} \quad (3.1)$$

4. Generalities on local terms

Suppose that X' is smooth over k and, in addition, that the number of fixed points of f is finite. It follows in particular that the number of fixed points of $g^{-1}f'$ is finite for every $g \in G$.

For each fixed point $x \in X$ of f , introduce the φ -central function

$$S_x(f', f, G, \varphi) : G \longrightarrow \mathbb{Q},$$

with rational values, defined by the formula

$$S_x(f', f, G, \varphi)(g) = \frac{1}{c^\varphi(g)} \sum_{x' \in X'_x{}^{g^{-1}f'}} (\nu_{x'}(g^{-1}f') - 1). \quad (4.1)$$

We have used the following notation:

$$c^\varphi(g) = \text{card } G_g^\varphi, \quad G_g^\varphi = \{x \in G \mid \varphi(x)gx^{-1} = g\}, \quad (4.2)$$

$$X'_x{}^{g^{-1}f'} = \{x' \in X' \mid p(x') = x, (g^{-1}f')(x') = x'\}, \quad (4.3)$$

$$\nu_{x'}(g^{-1}f') = \text{the multiplicity of the fixed point } x' \text{ of } g^{-1}f'. \quad (4.4)$$

By composition with the canonical inclusions

$$\mathbb{Q} \hookrightarrow \mathbb{Q}_\ell,$$

the function $S_x(f', f, G, \varphi)$ gives φ -central functions $S_{x,\ell}(f', f, G, \varphi)$ on G with values in the field $\mathbb{Q}_\ell$ of ℓ -adic numbers.

Conjecture 4.5. *For each $\ell \neq \text{car}(k)$, the values of the function $S_{x,\ell}(f', f, G, \varphi)$ are ℓ -adic integers.*

We shall prove below that Conjecture 4.5 is true when the dimension of X is equal to 1.

Proposition 4.6. *Let R be a discrete valuation ring, $\mathfrak{m}$ its maximal ideal, π a uniformizer, $k = R/\mathfrak{m}$ its residue field, assumed algebraically closed, v its valuation, $f'_R : R \rightarrow R$ an endomorphism of R , and G a finite group of automorphisms of R . Suppose that the endomorphism f'_R and the automorphisms g_R induce the identity on k , and suppose in addition that an endomorphism of G is given,*

$$\varphi : G \longrightarrow G,$$

satisfying

$$f'_R \circ g_R = \varphi(g)_R \circ f'_R.$$

If, for every $g \in G$, one writes

$$\nu(g_R^{-1} f'_R) = v(g_R^{-1} f'_R(\pi) - \pi),$$

and if $c^\varphi(g)$ (cf. 4.2) is decomposed in the form

$$c^\varphi(g) = p^s q, \quad (p, q) = 1, \quad p = \text{car}(k),$$

then $q \mid (\nu(g_R^{-1} f'_R) - 1)$.

To prove Proposition 4.6 one reduces to the case $g = e$ and $\varphi = \text{id}_G$. Indeed one first reduces to the case $g = e$ by associating with the situation (f'_R, G, g, φ) the situation

$$(f''_R = g_R^{-1} f'_R, G, e, \varphi' = \text{int}(g) \circ \varphi),$$

where $\varphi'(a) = g^{-1} \varphi(a) g$, and by observing easily that

$$\nu(g_R^{-1} f') - 1 = \nu(f''_R) - 1$$

and

$$c^\varphi(g) = c^{\varphi'}(e) \quad (G_g^\varphi = G_e^{\varphi'}) \quad (\text{cf. 4.2}).$$

To reduce simultaneously to the case $g = e$, $\varphi = \text{id}_G$, note that one can associate with the situation (f''_R, G, e, φ') the situation

$$(f''_R, G_e^{\varphi'}, e, \text{id}_{G_e^{\varphi'}}),$$

and that

$$G_e^{\varphi'} = (G_e^{\varphi'})_e^{\text{id}_{G_e^{\varphi'}}}.$$

The proof of Proposition 4.6 essentially uses the following lemma.

Lemma 4.7. *With the hypotheses and notation of Proposition 4.6, suppose moreover that $\varphi = \text{id}_G$, that is, that the operations of G on R commute with f'_R , and that $r = \nu(f'_R) \geq 2$. Let*

$$c = \text{card}(\text{Im}(G \longrightarrow \text{Aut}_k(\mathfrak{m}/\mathfrak{m}^2))).$$

Then the following divisibility relation holds:

$$c \mid (r - 1).$$

Proof. It is clear that the integer r satisfies the following conditions:

a) the homomorphism

$$f'_r : A/\mathfrak{m}^r \longrightarrow A/\mathfrak{m}^r$$

induced by f'_R is the identity;

b) the homomorphism

$$f'_{r+1} : A/\mathfrak{m}^{r+1} \longrightarrow A/\mathfrak{m}^{r+1}$$

is different from the identity.

On the other hand, it is evident that G acts on all the vector spaces $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ over k .

We first show that under the hypotheses of the lemma there exists a unique homomorphism of $k[G]$ -modules

$$u(f'_R) : \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \mathfrak{m}^r/\mathfrak{m}^{r+1}$$

satisfying the two following conditions:

- 1) $u(f'_R) \neq 0$;
- 2) $u(f'_R)(x \bmod \mathfrak{m}^2) = (f'_R(x) - x) \bmod \mathfrak{m}^{r+1}$, for every $x \in \mathfrak{m}$.

It is clear that condition a) implies $f'_R(x) - x \in \mathfrak{m}^r$ for every $x \in R$, which gives meaning to condition 2). This observation also gives the possibility of defining a homomorphism of abelian groups

$$\mathfrak{m} \longrightarrow \mathfrak{m}^r/\mathfrak{m}^{r+1}, \quad x \longmapsto (f'_R(x) - x) \bmod \mathfrak{m}^{r+1}.$$

This homomorphism vanishes on $\mathfrak{m}^2$. Indeed, if $x_1, x_2 \in \mathfrak{m}$, then $f'_R(x_1x_2) - x_1x_2 \in \mathfrak{m}^{r+1}$ because, by condition a),

$$\begin{aligned} f'_R(x_1) &= x_1 + \alpha_1, & \alpha_1 &\in \mathfrak{m}^r, \\ f'_R(x_2) &= x_2 + \alpha_2, & \alpha_2 &\in \mathfrak{m}^r, \end{aligned}$$

and hence

$$f'_R(x_1x_2) = f'_R(x_1)f'_R(x_2) = (x_1 + \alpha_1)(x_2 + \alpha_2) = x_1x_2 + x_1\alpha_2 + x_2\alpha_1 + \alpha_1\alpha_2,$$

and, plainly, $x_1\alpha_2 + x_2\alpha_1 + \alpha_1\alpha_2 \in \mathfrak{m}^{r+1}$.

It follows by passage to the quotient that we get a homomorphism of abelian groups

$$u(f'_R) : \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \mathfrak{m}^r/\mathfrak{m}^{r+1}.$$

The homomorphism $u(f'_R)$ is in fact a homomorphism of vector spaces over k . To see this, one must show that for all $x \in \mathfrak{m}$, $\alpha \in R$,

$$f'_R(\alpha x) - \alpha x \equiv \alpha(f'_R(x) - x) \pmod{\mathfrak{m}^{r+1}},$$

which follows immediately from the fact that f'_R induces the identity on k .

If the vector space

$$\mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1})$$

is endowed with its usual G -module structure, then its element $u(f'_R)$ belongs to the subspace

$$\mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1})^G$$

of G -invariants.

On the other hand, it is classical that the vector space $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ identifies canonically with the i -th tensor power $\bigotimes_k^i(\mathfrak{m}/\mathfrak{m}^2)$; consequently the vector space $\mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1})$ identifies, by the homomorphism of “contraction with the $(r-1)$ -st tensor power $\bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2)$ ”,

$$\gamma : \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1}) \longrightarrow \bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2).$$

Through the isomorphism γ , the G -module structure of $\mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, \mathfrak{m}^r/\mathfrak{m}^{r+1})$ is transported to the G -module structure of $\bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2)$ induced by the G -module structure of $\mathfrak{m}/\mathfrak{m}^2$:

$$g(\xi_1 \otimes \xi_2 \otimes \cdots \otimes \xi_{r-1}) = g\xi_1 \otimes \cdots \otimes g\xi_{r-1}.$$

We denote by $\mu(f'_R)$ the image of $u(f'_R)$ under the isomorphism γ . Thus

$$1) \ 0 \neq \mu(f'_R) \in \bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2)^G,$$

and it is clear that condition b) implies that

$$2) \ \mu(f'_R) \text{ belongs to no subspace of } \bigotimes_k^{r-1}(\mathfrak{m}/\mathfrak{m}^2) \text{ of the form}$$

$$\bigotimes_k^{r-1}(V), \quad V \subsetneq \mathfrak{m}/\mathfrak{m}^2.$$

Suppose, for contradiction, that simultaneously

$$f'_R(x) - x \in \mathfrak{m}^{r+1}, \quad f'_R(x') - x' \in \mathfrak{m}^{r+1}.$$

Then, in the field of fractions K of A , we have

$$\frac{f'_R(x)}{x} \in 1 + \mathfrak{m}^r, \quad \frac{f'_R(x')}{x'} \in 1 + \mathfrak{m}^r.$$

We may suppose that $f'_R(x') \neq 0$, since otherwise x' is the desired element. Under this hypothesis, using the fact that $1 + \mathfrak{m}^r$ is a multiplicative group, it follows that

$$\frac{x'}{f'_R(x')} \in 1 + \mathfrak{m}^r,$$

and consequently

$$\frac{f'_R(x)}{x} \cdot \frac{x'}{f'_R(x')} \in 1 + \mathfrak{m}^r.$$

But this last relation implies

$$\frac{f'_R(x_0)}{x_0} \in 1 + \mathfrak{m}^r.$$

Indeed,

$$\frac{f'_R(x_0)}{x_0} - \frac{f'_R(x)}{x} \cdot \frac{x'}{f'_R(x')} = \frac{f'_R(\frac{x}{x'})}{\frac{x}{x'}} - \frac{f'_R(x)}{x} \cdot \frac{x'}{f'_R(x')} = \frac{x'f'_R(\frac{x}{x'}) - x'f'_R(x)}{x} = 0.$$

Thus $f'_R(x_0) - x_0 \in \mathfrak{m}^{r+1}$, a contradiction.

Therefore $c \mid (r - 1)$, as was required.

Proof of Proposition 4.6. We have already noted that one may reduce to the case $g = e$, $\varphi = \text{id}_G$. If G_1 denotes the kernel of the homomorphism

$$G \longrightarrow \text{Aut}_k(\mathfrak{m}/\mathfrak{m}^2),$$

then ramification theory gives

$$\text{card}(G_1) = p^t.$$

Consequently, with the notation of Lemma 4.7,

$$\text{card}(G) = p^t c.$$

Since, on the other hand, $c^{\text{id}_G}(e) = \text{card}(G)$, the assertion of Proposition 4.6 in the case $g = e$, $\varphi = \text{id}_G$, and hence in general, follows from Lemma 4.7.

Corollary 4.8. *For each $\ell \neq \text{car}(k)$, the image of the rational number*

$$\frac{\nu(g_R^{-1} f'_R) - 1}{c^\varphi(g)}$$

under the canonical inclusion

$$\mathbb{Q} \hookrightarrow \mathbb{Q}_\ell$$

is an ℓ -adic integer.

Proposition 4.9. *Conjecture 4.5 is true if the dimension of the scheme X is equal to 1.*

Proof. We introduce the endomorphism

$$\psi : G \longrightarrow G$$

defined by

$$\psi = \text{int}_g \circ \varphi, \quad \text{int}_g(\xi) = g^{-1} \xi g.$$

It is easy to see that:

- i) $\psi(G_{x'}) \subset G_{g^{-1}f'x'}$, where $G_{x'}$ is the stabilizer of the point x' ;
- ii) $c^\varphi(g) = \text{card}(G^\psi)$, where $G^\psi = \{\xi \mid \psi(\xi) = \xi\}$;
- iii) G^ψ acts on $X'_x{}^{g^{-1}f'}$;

- iv) $\nu_{x'}(g^{-1}f') = \nu_{hx'}(g^{-1}f')$ for every $h \in G^\psi$;
- v) $(g^{-1}f')(g_1x) = \psi(g_1)(g^{-1}f')(x)$.

Consequently one can decompose $X'_x{}^{g^{-1}f'}$ as the union of its orbits and obtain

$$\sum_{x' \in X'_x{}^{g^{-1}f'}} (\nu_{x'}(g^{-1}f') - 1) = \sum_{x' \in X'_x{}^{g^{-1}f'} / G^\psi} \frac{\nu_{x'}(g^{-1}f') - 1}{\text{card}(G^\psi_{x'})} \text{card}(G^\psi), \quad (4.9.1)$$

and Proposition 4.9 follows from Corollary 4.8 and formula (4.9.1).

5. Generalization of the Nielsen–Wecken formula

5.1. Theorem

Suppose that the scheme X' is proper and smooth over k , and suppose moreover that the number of fixed points of f is finite. Then one has the following formula:

$$\text{Tr}_{\mathbb{Z}_\ell[G], \varphi}^*(f_{H_{U', X'}^\vee}^\vee)(g) = \sum_{x \in X^f} \nu_x(f) \text{NW}_f^{G, \varphi}(x)(g) - \sum_{\substack{x \in Y^f \\ x' \in X'_x{}^{g^{-1}f'}}} \frac{1 - \nu_{x'}(g^{-1}f')}{c^\varphi(g)} \quad (5.1.1)$$

for every $g \in G$.

Proof. First, by reducing modulo ℓ^n for every $n \geq 1$ and using (XI, 7.2.3.6), one sees that for every $g \in G$,

$$c^\varphi(g) \text{Tr}_{\mathbb{Z}_\ell[G], \varphi}(f_{H_{U', X'}^\vee}^\vee)(g) = \text{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{H_{U', X'}^\vee}(\ell). \quad (5.2)$$

The commutative diagram (1.3), whose rows are exact, gives (XI, §4)

$$\text{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{H_{U', X'}^\vee}(\ell) = \text{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{R\Gamma_{X'}(\mathbb{Z}_\ell, X')} - \text{Tr}_{\mathbb{Z}_\ell}^*(g_{Y'}^{-1}f'_{Y'})_{R\Gamma_{Y'}(\mathbb{Z}_\ell, Y')}. \quad (5.3)$$

The two terms on the right-hand side of (5.3) can be evaluated by the classical Lefschetz formula. For the last term this is in fact a trivial Lefschetz formula:

$$\text{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{R\Gamma_{X'}(\mathbb{Z}_\ell, X')} = \sum_{x \in X^f} \sum_{x' \in X'_x{}^{g^{-1}f'}} \nu_{x'}(g^{-1}f'), \quad (5.4)$$

$$\text{Tr}_{\mathbb{Z}_\ell}^*(g_{Y'}^{-1}f'_{Y'})_{R\Gamma_{Y'}(\mathbb{Z}_\ell, Y')} = \text{the number of fixed points of } g_{Y'}^{-1}f'_{Y'}. \quad (5.5)$$

Consequently

$$\text{Tr}_{\mathbb{Z}_\ell}(g_{X'}^{-1}f')_{H_{U', X'}^\vee}(\ell) = \sum_{x \in U^f} \sum_{x' \in X'_x{}^{g^{-1}f'}} \nu_{x'}(g^{-1}f') - \sum_{x \in Y^f} \sum_{x' \in X'_x{}^{g^{-1}f'}} (1 - \nu_{x'}(g^{-1}f')). \quad (5.6)$$

But, for a given $x \in U^f$, the set $X'_x{}^{g^{-1}f'}$ is non-empty if and only if $\text{NW}_{f'}^{G, \varphi}(x) = 1$. Moreover one has

$$\text{card } X'_x{}^{g^{-1}f'} = c^\varphi(g), \quad x \in U, \quad (5.7)$$

$$\nu_{x'}(g_{X'}^{-1}f') = \nu_x(f), \quad x \in U. \quad (5.8)$$

The relations (5.2) and (5.6) therefore give

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G], \varphi}(f_{H_{U', X'}(\ell)}^{\vee})(g) = \sum_{x \in U^f} \nu_x(f) \mathrm{NW}_f^{G, \varphi}(x)(g) - \sum_{x \in Y^f} \sum_{x' \in X'_x g^{-1} f'} \frac{1 - \nu_{x'}(g_{X'}^{-1}f')}{c^\varphi(g)}, \quad (5.9)$$

which is the formula to be proved.

5.10. Special cases

a) The unramified case: $Y' = \emptyset$, that is, $Y = \emptyset$.

In this case formula (5.1.1) becomes

$$\mathrm{Tr}_{\mathbb{Z}_\ell[G], \varphi}(f_{H_{U', X'}(\ell)}^{\vee}) = \sum_{x \in X^f} \nu_x(f) \mathrm{NW}_f^{G, \varphi}(x). \quad (5.11)$$

b) The case of multiplicities 1: $\nu_{x'}(g_{X'}^{-1}f') = 1$ for every $x' \in Y' \cap X'_x g^{-1}f'$ and every $g \in G$.

In this case formula 5.1.1 reduces to formula 5.11. This case occurs in particular if $\dim X = 1$ and $\nu_x(f) = 1$ for every $y \in Y$.

6. Application to a Lefschetz formula

6.1

We keep the notation and the conditions of number 1, except where the contrary is expressly stated. Let A be a commutative noetherian Λ_n -algebra, let F be a constructible sheaf, for the étale topology, of perfect A -modules on X , and let

$$u : f^*(F) \longrightarrow F$$

be a homomorphism of sheaves of A -modules.

In particular, one obtains an endomorphism of the pair (X, F) :

$$(u, f) : (X, F) \longrightarrow (X, F),$$

and consequently an endomorphism of complexes of A -modules:

$$R\Gamma_X(u, f) : R\Gamma_X(F) \longrightarrow R\Gamma_X(F).$$

Since the scheme X is proper over k by hypothesis and the sheaf F is constructible and torsion, the finiteness theorem (SGA 4, XVII 5) ensures that the complex $R\Gamma_X(F)$ is a perfect complex of A -modules. The trace

$$\mathrm{Tr}_A(R\Gamma_X(u, f))$$

is therefore defined, with

$$\mathrm{Tr}_A(R\Gamma_X(u, f)) \in A.$$

Our aim will be to give, under some additional hypotheses, an explicit formula for $\text{Tr}_A(R\Gamma_X(u, f))$ in terms of local invariants associated with the fixed points of f .

The notation $x \in X^f$ denotes the set of closed fixed points of X .

Suppose that an endomorphism $f' : X' \rightarrow X'$ and an endomorphism $\varphi : G \rightarrow G$ of the group G are given, satisfying the conditions of number 1. Suppose moreover that the sheaf F becomes constant on U' (or at least that $F|U$ is trivialized by the covering $U' \rightarrow U$). Then one has the formula

$$\text{Tr}_A(R\Gamma_X(u, f)) = \sum_{x \in X^f} \tau_x(f, u, f', G, \varphi), \quad (6.2)$$

where

$$\tau_x(f, u, f', G, \varphi) = S_{x, \ell^n}(f, f', G, \varphi)(e) \cdot \text{Tr}_A(u_x) \quad (6.3)$$

and

$$\alpha_x(f, u, f', G, \varphi) = S_{x, \ell^n}(f, f', G, \varphi)(e) \cdot \text{Tr}_A(u_M^\vee). \quad (6.4)$$

It follows, taking into account the relation $X^f = Y^f$, that

$$\text{Tr}_{\Lambda_n[G], \varphi}(f_{H_{n, U', X'}}^{\vee})(g) = \sum_{x \in X^f} S_{x, \ell^n}(f, f', G, \varphi)(g). \quad (6.3.12)$$

Combining the relations 6.3.12, 6.3.11, 6.3.5, 6.3.4 gives the formula of the statement.

6.4. Remarks on the local terms

a)

It is easy to see that, if $g \in G$, then one has the relation

$$\alpha_x(f, u, g^{-1}f', G, \varphi_g) = \alpha_x(f, u, f', G, \varphi), \quad (6.4.1)$$

where

$$\varphi_g(\xi) = (\text{int}_g \circ \varphi)(\xi) = g^{-1}\varphi(\xi)g, \quad \forall \xi \in G. \quad (6.4.2)$$

Relation 6.4.1 allows us always to reduce, after replacing f' by $g^{-1}f'$ and φ by φ_g , to the case where there exists at least one point $x' \in X_x'^{f'}$.

b)

Let $x' = X_x'^{f'}$ and let $G_{x'}$ be the stabilizer of the point x' . Evidently $G_{x'}$ is stable under φ , and hence one obtains an endomorphism

$$\varphi_{x'} = \varphi|_{G_{x'}} : G_{x'} \longrightarrow G_{x'}.$$

Let $X'/G_{x'}$ denote the quotient scheme obtained from X' by the equivalence relation defined by $G_{x'}$, let $F'/G_{x'}$ be the endomorphism of $X'/G_{x'}$ obtained from the endomorphism f' of X' by the same equivalence relation, and let x' denote the class of the point x' modulo this equivalence relation.

The canonical projection

$$p : X' \longrightarrow X'/G = X$$

factors as

$$p : X' \xrightarrow{p_1} X'/G_{x'} \xrightarrow{p_2} X = X'/G.$$

Taking the inverse image by p_2 of the homomorphism ν , one obtains a homomorphism

$$\nu = p_2(u) : (f'/G_{x'})^*(p_2^*(F)) \longrightarrow p_2^*(F).$$

It is easy to see that

$$\alpha_{x'}(f'/G_{x'}, \nu, f', G_{x'}, \varphi_{x'}) = \alpha_x(f, u, f', G, \varphi) \quad (x = p(x')). \quad (6.4.3)$$

Thus one may always reduce to the “totally ramified” case, that is, to the case where the fibre of the point x is reduced to one point.

c) The “tamely ramified” case.

Suppose that there exists $x' \in p^{-1}(x)$ such that

$$f'(x') = x', \quad p(x') = x,$$

a case to which one can always reduce, cf. case a), and suppose that the order of the group $G_{x'}$ is prime to the characteristic of the field k . Then one has

$$\alpha_x(f, u, f', G, \varphi) = S_{x, \ell^n}(f, f', G, \varphi)(e) \operatorname{Tr}_A(u_M^\vee). \quad (6.4.4)$$

Taking case b) into account, one may suppose that $G = G_{x'}$. Formula (6.4.4) then follows from the following lemma.

Lemma 6.4.5

If the following relations hold:

$$\begin{aligned} \nu_{x'}(f') &> 1, \\ \nu_{x'}(g^{-1}f') &> 1, \end{aligned}$$

then g belongs to the first ramification group G_1 (cf. 4.7):

$$g \in G_1,$$

that is, g induces the identity on the vector space $\mathfrak{m}_{x'}/\mathfrak{m}_{x'}^2$.

Proof

Let $\widehat{\mathcal{O}}_{x'} = k[[T]]$. The automorphism of $\mathfrak{m}/\mathfrak{m}^2$ induced by g^{-1} is determined by a root of unity $\varepsilon(g^{-1}) \in k$ by the condition

$$g^{-1}T = \varepsilon(g^{-1})T \pmod{\mathfrak{m}^2},$$

where $\mathfrak{m}$ is the maximal ideal of the ring $k[[T]]$.

On the other hand, the hypothesis $\nu_{x'}(f') > 1$ gives

$$f'(T) = T + a_\nu T^\nu + a_{\nu+1} T^{\nu+1} + \cdots, \quad \nu \geq 2.$$

Consequently

$$g^{-1} f'(T) = \varepsilon(g^{-1}) T + b_\nu T^\nu + b_{\nu+1} T^{\nu+1} + \cdots,$$

and the hypothesis $\nu(g^{-1} f') > 1$ implies $\varepsilon(g^{-1}) = 1$, as required.

d) The case of a sheaf F which is not ramified at the fixed points of the endomorphism f .

In this case the local term $\tau_x(f, u, f', G, \varphi)$ is given by the formula

$$\tau_x(f, u, f', G, \varphi) = \nu_x(f) \operatorname{Tr}_A(u_x), \quad (6.4.6)$$

and consequently depends only on the pair (f, u) .

Indeed, in this case one has

$$S'_{x, \ell^n}(f, f', G, \varphi)(g) = \begin{cases} 0 & \text{if there exists } x' \in X' \text{ such that } (g^{-1} f')(x') = x' \text{ or} \\ & \text{equivalently if the } \varphi\text{-conjugacy class of } g \text{ coincides with} \\ & \text{the class } \operatorname{NW}_f^{G, \varphi}(x) \text{ (cf. 3);} \\ \nu_x(f) - 1 & \text{otherwise.} \end{cases} \quad (6.4.7)$$

It follows in particular that

$$S_{x, \ell^n}(f, f'; G, \varphi)(g) = 0 \quad \text{if } g \notin \operatorname{NW}_f^{G, \varphi}(x). \quad (6.4.8)$$

We therefore find

$$\begin{aligned} \tau_x(f, u, f', G, \varphi) &= \nu_x(f) \operatorname{Tr}_A(g^{-1} u_M^\vee) = \nu_x(f) \operatorname{Tr}_A(g^{-1} u')_{x'} \\ &= \nu_x(f) \operatorname{Tr}_A(u_x). \end{aligned} \quad (6.4.9)$$

7. Comments on the conditions of validity of the Lefschetz formula

7.1

Of course, in practice one starts with a scheme X proper over k and an endomorphism

$$(u, f) : (X, F) \longrightarrow (X, F),$$

such that:

- F is a constructible sheaf of A -modules,
- the set X^f of fixed points of f is finite.

The problem is to define, under these conditions, local terms $\tau_x(u, f)$, $x \in X^f$, such that

$$(*) \quad \mathrm{Tr}_A(R\Gamma_X(u, f)) = \sum_{x \in X^f} \tau_x(u, f).$$

Using the two duality theorems, global and local, Verdier succeeds in defining terms $\tau_x(u, f)$ such that formula (*) is true (cf. III). Unfortunately, the explicit determination of the $\tau_x(u, f)$, as defined by Verdier, in terms of known local invariants seems difficult; hence the interest of the definitions proposed in the preceding chapter.^(*)

By contrast, these latter definitions require additional hypotheses on F , f , and u , namely the existence of U, U', G, f', φ having the properties required in 5.1, 6.1, 6.2.

^(*) See, however, III B for the comparison of the local terms of 6.3.1 with Verdier's local terms.